Boundary-Guided Replay: Learning at the Success–Failure Boundary of Decision Policies

Anonymous submission

Abstract

Policies trained on nominal starts often fail under small deviations from the states encountered during training. Existing curriculum and level-replay methods prioritize difficult or high-regret levels, but they do not explicitly identify where a policy's behavior transitions from likely success to likely failure. We introduce Boundary-Guided Replay (BGR), a replay method for resettable or replayable sequential decision settings. For each replayable decision state, BGR estimates a recovery curve measuring success probability under perturbations of increasing scale, derives a critical radius at which performance crosses a success threshold, and samples training perturbations near this state-conditioned boundary. To make boundary estimation practical, BGR uses active probing, monotone curve fitting, uncertainty-aware refresh, and diversity-preserving replay. In synthetic and procedural grid-margin benchmarks, BGR improves recovery AUC, clean success, and learning-curve area over uniform replay; the grid full-baseline comparison and held-out replication pool to 60/0 paired RAUC wins over uniform (0.4342 vs. 0.3965); in a robot-suffix simulator, a coverage-aware BGR variant improves final object recovery AUC (0.4971 vs. 0.4856), clean success, transfer recovery AUC, and learning-curve area across 60 paired seeds, while still trailing uniform on median critical radius. Fixed-radius, failure-focused, and loss-priority baselines underperform in the controlled sweeps. OpenVLA/LIBERO audits show that the same recovery-margin interface can measure learned-policy brittleness and build matched fine-tuning datasets; because current adaptation does not establish stable gains over the official checkpoint, we report them as audits, not as final robotics fine-tuning claims.

1 Introduction

A policy can solve a task and still be brittle: a minor shift in object pose, grid layout, browser state, or trajectory prefix may move it outside its recovery basin. Standard success metrics hide this geometry. We ask a different question: for each replayable state, how far can the current policy be perturbed before success becomes failure?

Existing replay and curriculum methods rank examples by difficulty, temporal-difference error, regret, novelty, or failure. These signals are useful, but they do not explicitly model the local response curve around a state. BGR instead estimates a state-conditioned recovery curve and trains near its success–failure transition.

Our contributions are:

- a domain-agnostic replay interface based on replayable decision states, perturbation families, recovery curves, critical radii, and recovery AUC;
- an efficient BGR algorithm that actively estimates success–failure boundaries and samples both states and perturbation radii near those boundaries;
- an active-estimation validation showing that boundary-focused probes recover critical radii with small fixed budgets rather than dense sweeps;
- synthetic, procedural grid-margin, and robot-suffix simulator benchmarks showing where boundary replay and coverage-aware boundary replay expand recovery margins under paired-seed comparisons, including held-out 60-seed pooled evidence for the grid and suffix studies;
- OpenVLA/LIBERO audits that define the practical scope of the current learned-policy evidence and separate recovery-margin measurement from robotics fine-tuning claims.

2 Related Work

BGR is closest in spirit to curriculum learning and level replay for reinforcement learning. Prioritized Level Replay samples levels by learning-potential proxies such as value loss (Jiang, Grefenstette, and Rocktäschel 2021), while PAIRED and ACCEL generate or mutate environments using regret and curriculum-frontier objectives (Dennis et al. 2020; Parker-Holder et al. 2022). BGR is complementary: it can operate over fixed replayable states and prioritizes the local perturbation geometry of the current policy rather than a scalar difficulty or regret score. It also differs from broad curriculum schedules surveyed by Narvekar et al. (2020), because the unit of progress is recovery-margin expansion around each state.

Prioritized experience replay ranks transitions by temporal-difference error (Schaul et al. 2016), and active learning selects informative labels under a query budget (Settles 2009). BGR borrows the budgeted-query spirit but changes the target: it actively estimates a success-probability curve around a replay state, then trains at a state-conditioned perturbation radius. It is also distinct from adversarial-example training (Goodfellow, Shlens, and Szegedy 2015; Madry et al. 2018): BGR is not an adversarial-training objective and does not solve a worst-case inner maximization; it samples feasible perturbation scales near the observed recovery boundary to expand a policy's local margin.

BGR also connects to robustness training and robotics generalization. Domain randomization improves transfer by broadening training distributions (Tobin et al. 2017), but fixed perturbation distributions can oversample states that are already easy or currently infeasible. BGR instead estimates where each state's perturbation curve crosses from success to failure and trains near that radius. Our procedural experiments use resettable grid states in the style of lightweight grid benchmarks (Chevalier-Boisvert et al. 2023); the suffix study instantiates the same interface for manipulation-style states related to LIBERO (Liu et al. 2023) and vision-language-action policies such as OpenVLA (Kim et al. 2024).

3 Problem Setting

Let π_θ be a sequential decision policy and let $Y(\tau) \in \{0,1\}$ denote terminal success for rollout τ. A replayable decision state $x \in \mathcal{X}$ is any object from which evaluation or training can restart, such as an environment seed, simulator checkpoint, task prefix, or robot suffix state.

For each x, a domain-valid perturbation family applies $\tilde{x} = T(x, \delta)$ with $\delta \sim \mathcal{P}_x(\sigma)$, where $\sigma \geq 0$ is a perturbation magnitude. The perturbation family is part of the benchmark definition: grid perturbations preserve free cells and shortest-path reachability, suffix perturbations preserve teacher feasibility, and OpenVLA observation perturbations preserve the task instruction and simulator state while modifying the visual input stream. The recovery curve is

$$R_\pi(x, \sigma) = \Pr_{\delta,\tau}[Y(\tau) = 1 \mid \tau \sim \pi(\cdot \mid T(x, \delta))]. \quad (1)$$

For threshold α, the critical radius is

$$r_{\pi,\alpha}(x) = \sup\{\sigma : R_\pi(x, \sigma) \geq \alpha R_\pi(x, 0)\}. \quad (2)$$

We evaluate robustness with recovery AUC,

$$\mathrm{RAUC}_\pi(x) = \frac{1}{\sigma_{\max}} \int_0^{\sigma_{\max}} R_\pi(x, \sigma) d\sigma. \quad (3)$$

BGR also assumes a feasibility witness $F(x, \sigma)$ that estimates whether perturbed states remain meaningful and solvable by a reference process. In our experiments this witness is either an exact reachability check, a teacher/demo suffix model, a simulator validity check, or observed clean OpenVLA success. The witness is not an oracle for the learned policy; it filters invalid or impossible perturbations so replay focuses on recoverable boundaries rather than broken tasks.

4 Boundary-Guided Replay

BGR maintains a buffer of replayable states. For each state, it probes perturbation radii, fits a monotone recovery curve, estimates a critical radius, and assigns replay priority to states whose boundaries are feasible, sharp, uncertain, and near a trainable target band. The method therefore separates three decisions: whether a state is valid enough to replay, which replay state is currently informative, and which perturbation radius lies near that state's success–failure transition.

Study	**Replay state**	**Perturbation**	**Witness/eval**
Synthetic	latent recovery state	scalar radius	dense curve
Grid-margin	mid-path grid state	valid radius shifts	reachability
Robot suffix	demo suffix phase	object/EEF offsets	teacher feasibility
OpenVLA audit	LIBERO state	object/visual changes	simulator success

Table 1: Instantiations of the BGR interface. The first three rows are training experiments; the OpenVLA row is an audit of recovery curves and data plumbing, not a fine-tuning claim.

Boundary-Guided Replay
Initialize replay buffer $\mathcal{B}$ with replayable states.
For each $x \in \mathcal{B}$, probe $\sigma \in \{0, \sigma_{\min}, \sigma_{\mathrm{mid}}, \sigma_{\max}\}$, fit monotone $\hat{R}(x, \sigma)$, and estimate $\hat{r}_\alpha(x)$.
For each training iteration, sample $x \sim q_{\mathcal{B}}(x)$ proportional to BGR priority, sample σ from a mixture centered at $\hat{r}_\alpha(x)$, apply $\delta \sim \mathcal{P}_x(\sigma)$, collect rollout or teacher recovery data, update π_θ, and periodically refresh stale curve estimates.

Method box: BGR training loop without domain-specific learner details.

The priority for a record is a product of gates and soft utilities:

$$P(x) = G(x) U_{\mathrm{boundary}}(x) U_{\mathrm{sharp}}(x)^\lambda U_{\mathrm{unc}}(x)^\beta U_{\mathrm{stale}}(x)^\eta. \quad (4)$$

Here $G(x)$ gates out states with low clean success or low witness feasibility, U_{boundary} prefers critical radii in a trainable band, U_{sharp} prefers informative transitions, U_{unc} refreshes uncertain curve estimates, and U_{stale} revisits old records as the policy changes. We mix this distribution with uniform replay to prevent starvation. For the radius-level decision, BGR samples from a mixture centered at $\hat{r}_\alpha(x)$ with clean, easier, harder, and broad-radius coverage components; ablations test which part is responsible for the gains.

The mechanism is simple under a local margin-update model.

Proposition 1. Fix a replay state x and let $r = r_{\pi,\alpha}(x)$. Suppose training at radius σ gives an expected first-order RAUC gain $k_x(|\sigma - r|)$ for a nonincreasing kernel k_x. If a boundary sampler $p_b(\sigma \mid x)$ induces a distance-to-boundary random variable $D_b = |\sigma - r|$ satisfying $\Pr[D_b > t] \leq \Pr[D_0 > t]$ for every t relative to a baseline sampler's distance D_0, then

$$\mathbb{E}_{p_b}[k_x(|\sigma - r|)] \geq \mathbb{E}_{p_0}[k_x(|\sigma - r|)]. \quad (5)$$

The assumption is first-order stochastic dominance over distance to the boundary. By the quantile coupling for stochastic orders (Shaked and Shanthikumar 2007), there exists a coupling with $D_b' \leq D_0'$ almost surely and with the same marginals as D_b, D_0; monotonicity of k_x then gives $\mathbb{E}[k_x(D_b')] \geq \mathbb{E}[k_x(D_0')]$, which is the stated inequality. Thus, when recovery-margin updates are local, placing more probability near the current critical radius improves expected margin expansion over uniform or fixed-radius replay. BGR uses boundary-centered sampling as this local default, while retaining clean, easier, harder, broad-radius, and uniform components because real learners can violate the local model through saturation, noisy labels, or unstable optimization.

5 Experiments

All experiments are driven by artifact YAML configs and scripts. The anonymous artifact records per-seed CSVs, aggregation code, paired exact sign tests, Slurm launch commands, run ledger files, compute/GPU environment snapshots, and an MIT license; the paper tables are generated from these artifacts rather than hand-edited results. We use three evidence tiers. The main claim is supported by controlled synthetic and procedural grid-margin experiments where the recovery-margin object is directly trainable and baselines share paired seeds; the grid result is replicated on held-out seeds and pooled over 60 paired comparisons. The robot-suffix simulator is a manipulation-style extension: a coverage-aware BGR variant improves final object RAUC, clean success, transfer RAUC, and AULC over uniform suffix replay across original and held-out sweeps, while still trailing uniform on median critical radius. The OpenVLA/LIBERO results are learned-policy audits and infrastructure checks, not BGR fine-tuning claims.

5.1 Protocol and Metrics

We report clean success, recovery AUC (RAUC), median r_{80}, and area under the RAUC learning curve (AULC). RAUC and r_{80} are evaluated on dense held-out radius grids rather than on the probes used by BGR during training. Synthetic training, estimator validation, and packaged grid full-baseline, ablation, sensitivity, and suffix coverage confirmations use 30 paired seeds; grid learning-curve diagnostics use 15. We report exact two-sided paired sign tests; five-seed exploratory variants are reported only as exploratory evidence.

Baselines are matched to the decisions BGR changes. Uniform replay samples replay states and perturbation radii without recovery-boundary information. Fixed-radius replay uses the same learner and feasibility filters at one configured perturbation scale. Failure-only replay prioritizes states with observed failures without estimating a crossing radius. PLR-loss and loss-priority baselines replace BGR's recovery-curve priority with learner-loss proxies. Clean-only suffix training regularizes on unperturbed suffixes. Paired comparisons share seeds, rollout budgets, evaluation grids, and validity filters.

5.2 Active Boundary Estimation

BGR is useful only if recovery boundaries can be estimated without evaluating dense curves for every replay state. We therefore validate the training-time estimator separately from the replay experiments. For each synthetic state, we compute a dense 201-point reference curve for evaluation only, then compare 17-probe estimators: uniform random radii and active BGR probing initialized with a coarse sweep and then focused near the current critical-radius estimate.

Table 4 shows over 30 paired seeds that active BGR finds the boundary within ± 0.08 radius units more often than uniform probing (0.670 vs. 0.595) and lowers mean r_{80} error (0.081 vs. 0.105). Paired exact sign tests support both effects ($p < 10^{-3}$). This result does not claim that active probing dominates every fixed grid in this simple one-dimensional setting; rather, it verifies that BGR's boundary estimates are usable at a small fixed rollout budget before final dense-curve evaluation.

5.3 Synthetic Recovery-Margin Benchmark

The synthetic benchmark creates replayable states with latent feasible radii and policy recovery margins. Success probability follows a sigmoid recovery curve, and training updates near the current margin expand that margin most efficiently. This controlled environment tests whether BGR's estimator and sampler recover the intended behavior before higher-cost sequential or robotics runs.

In the 30-seed run, BGR improves final recovery AUC over uniform replay (0.3716 vs. 0.3633), improves the area under the RAUC learning curve (0.3149 vs. 0.2984), and preserves clean success (0.8899 vs. 0.8838), with exact sign tests supporting the RAUC and AULC comparisons (Table 3). Fixed-radius, failure-only, and PLR-loss proxy baselines underperform substantially in this controlled setting, suggesting that explicit boundary-radius sampling is useful when the benchmark contains learnable success–failure transitions.

5.4 Procedural Grid-Margin Study

We instantiate replayable states in procedural grid environments with resettable seeds and mid-path states. Generated obstacle maps define feasible perturbation neighborhoods through shortest-path reachability, while each replay state has a policy-specific recovery margin. This benchmark isolates whether BGR can expand state-conditioned recovery curves in a non-robotic sequential decision setting.

In a completed 30-seed paired grid-margin full-baseline run, BGR outperforms uniform replay on final RAUC (0.4344 vs. 0.3963), RAUC learning-curve area (0.3526 vs. 0.3132), median r_{80} (0.3447 vs. 0.3321), and clean success (0.9455 vs. 0.8939), with paired exact sign tests supporting the RAUC comparison (Table 3). A held-out seeds 30–59 BGR-vs-uniform replication gives 0.4340 vs. 0.3967 RAUC, also with 30/0 wins. Pooling original and held-out grid sweeps gives 0.4342 vs. 0.3965 RAUC with 60/0 paired wins. The same sweep shows that failure-only replay (RAUC 0.2909), PLR-loss proxy replay (0.2109), and fixed-radius perturbation training (0.2097) substantially underperform BGR (0.4344), with exact sign tests supporting all three RAUC deltas.

Stored 15-seed learning curves show BGR ahead of uniform on RAUC for all paired seeds at every post-update checkpoint (steps 30–300; $p = 0.0001$). The 30-seed sensitivity tables give the same pattern: target margins 0.26–0.54 yield RAUC 0.416–0.443 vs. uniform 0.396; obstacle regimes yield BGR RAUC 0.434–0.435 vs. uniform 0.396;

Benchmark	Method	Clean	RAUC	Median r_{80}	AULC
Synthetic	BGR	0.890±0.001	0.372±0.001	0.286±0.002	0.315±0.001
Synthetic	Uniform	0.884±0.001	0.363±0.001	0.282±0.002	0.298±0.001
Synthetic	Fixed	0.821±0.001	0.230±0.001	0.131±0.001	0.229±0.001
Synthetic	Failure	0.817±0.001	0.233±0.001	0.135±0.001	0.231±0.001
Synthetic	Loss-priority	0.819±0.001	0.229±0.001	0.134±0.001	0.229±0.001
GridMargin	BGR	0.945±0.000	0.434±0.000	0.345±0.001	0.353±0.000
GridMargin	Uniform	0.894±0.000	0.396±0.000	0.332±0.001	0.313±0.000
RobotSuffix	BGR-Coverage	0.864±0.001	0.497±0.001	0.498±0.001	0.382±0.001
RobotSuffix	Uniform	0.836±0.001	0.485±0.001	0.505±0.001	0.371±0.001

Table 2: Results generated from anonymous artifact CSVs. Synthetic, grid-margin full-baseline, and robot-suffix coverage comparisons use 30 paired seeds, with held-out 30-seed BGR-vs-uniform replications reported in text for grid and suffix. BGR is strongest in the synthetic and procedural grid-margin settings. In the lightweight robot-suffix simulator, the coverage-aware BGR-Suffix variant improves clean success, final object RAUC, transfer RAUC, and AULC over uniform suffix replay, but uniform retains a higher median critical radius.

Comparison	Metric	n	Mean diff.	95% CI	W/L/T	p
Synthetic vs uniform	final RAUC	30	0.0083	[0.0059, 0.0107]	29/1/0	< 0.0001
Grid vs uniform	final RAUC	30	0.0381	[0.0372, 0.0390]	30/0/0	< 0.0001
Suffix RAUC vs clean-only	final RAUC	30	0.2338	[0.2327, 0.2350]	30/0/0	< 0.0001
Suffix RAUC vs fixed	final RAUC	30	0.3386	[0.3373, 0.3399]	30/0/0	< 0.0001
Suffix RAUC vs failure-only	final RAUC	30	0.0741	[0.0731, 0.0750]	30/0/0	< 0.0001
Suffix RAUC vs loss-priority	final RAUC	30	0.3275	[0.3262, 0.3288]	30/0/0	< 0.0001
Suffix RAUC vs uniform	final RAUC	30	0.0116	[0.0109, 0.0122]	30/0/0	< 0.0001
Suffix transfer vs uniform	transfer RAUC	30	0.0061	[0.0054, 0.0068]	30/0/0	< 0.0001
Suffix AULC vs uniform	RAUC AULC	30	0.0115	[0.0108, 0.0122]	30/0/0	< 0.0001
Estimator 30-seed	boundary hit	30	0.0751	[0.0646, 0.0857]	30/0/0	< 0.0001

Table 3: Selected paired effect sizes for the central claims, including the completed 30-seed synthetic, estimator, grid, and suffix full-baseline confirmations. Mean differences and 95% confidence intervals are computed over paired seeds; W/L/T counts paired wins, losses, and ties in the declared improvement direction; p is the exact two-sided paired sign-test value. Synthetic, estimator, grid full-baseline, and suffix coverage-full comparisons use 30 paired seeds.

Estimator	r_{80} MAE	RAUC MAE	Hit rate
Active BGR	0.081±0.001	0.065±0.000	0.670±0.004
Uniform	0.105±0.001	0.066±0.000	0.595±0.004

Table 4: Probe-efficiency validation on synthetic recovery curves. Both methods use 17 Bernoulli probes per state and are evaluated against dense 201-point curves. Active BGR improves boundary-hit rate over uniform probing at the same rollout budget, while keeping critical-radius and RAUC error comparable.

and geometry stresses give stress RAUC 0.362–0.457 vs. uniform 0.323–0.430. The learning-rate table preserves the high-rate final-RAUC caveat. The reported 0.38 margin is not the best tuned value.

A learning-rate sweep is a scope diagnostic: BGR wins final RAUC at 0.015 and 0.030 (0.382 vs. 0.320; 0.434 vs. 0.396) and AULC at all three rates, but high learning rate flips final RAUC to uniform (0.485 vs. 0.491). Thus neither failure mining nor static perturbation scales recover the same margin-expansion behavior, while aggressive optimization can saturate uniform final RAUC. A separate tabular grid-policy setup either saturates under broad replay or undertrains BGR relative to fixed-radius oracle imitation; we therefore keep the main procedural claim on the margin-expansion benchmark.

Method	Clean	RAUC	Median r_{80}	AULC
BGR	0.945±0.000	0.434±0.000	0.345±0.001	0.353±0.000
Uniform	0.894±0.000	0.396±0.000	0.332±0.001	0.313±0.000
Failure-only	0.845±0.000	0.291±0.000	0.231±0.001	0.254±0.000
Loss-priority	0.797±0.001	0.211±0.000	0.137±0.001	0.213±0.000
Fixed radius	0.794±0.001	0.210±0.000	0.136±0.001	0.213±0.000

Table 5: Thirty-seed procedural grid-margin full-baseline comparison. Boundary-centered replay is the only method that both preserves clean success and expands recovery margins.

Method	Clean	RAUC	Median r_{80}	AULC
BGR	0.945±0.000	0.434±0.000	0.345±0.001	0.353±0.000
No uncertainty	0.946±0.000	0.435±0.000	0.344±0.001	0.353±0.000
No sharpness	0.946±0.000	0.435±0.000	0.344±0.001	0.353±0.000
Uniform radius	0.891±0.000	0.382±0.000	0.322±0.001	0.305±0.000
Uniform replay	0.894±0.000	0.396±0.000	0.332±0.001	0.313±0.000

Table 6: Grid-margin ablations over 30 seeds. Boundary-centered

radius sampling is the important ingredient in this benchmark: replacing it with uniform radii drops RAUC below uniform replay, while removing uncertainty or sharpness weights has little effect.

Table 6 isolates the replay decisions inside BGR. The uncertainty and sharpness factors are not decisive in this benchmark, but the radius-level boundary rule is: keeping BGR state priority while sampling perturbation radii uniformly reduces RAUC from 0.434 to 0.382 and AULC from 0.353 to 0.305, even below uniform replay's 0.396 RAUC. Paired exact sign tests support both the BGR-over-uniform-radius gap and the uniform-radius-below-uniform gap ($p = 0.0001$). This supports the central claim that BGR is not only hard-state prioritization; it also depends on training at the estimated success–failure radius.

5.5 Robot Suffix Study

BGR-Suffix treats a robot demonstration suffix as a replayable state. The lightweight suffix simulator models language-conditioned manipulation families, suffix phase, object-pose and end-effector perturbations, teacher feasibility, and clean-demo regularization. Training uses object-pose offsets; end-effector offsets are a perturbation-family transfer test.

The first 15-seed run showed that boundary-heavy BGR-Suffix undercovered high-radius recovery. The 30-seed coverage-aware run fixes this: BGR-Coverage raises final object RAUC over uniform (0.4969 vs. 0.4854), failure-only (0.4229), clean-only (0.2631), loss-priority (0.1694), and fixed-radius replay (0.1583), with 30/0 paired wins. A held-out seeds 30–59 replication gives 0.4972 vs. 0.4859 RAUC, also with 30/0 wins. Pooling original and held-out suffix sweeps gives object RAUC 0.4971 vs. 0.4856, clean 0.8642 vs. 0.8367, transfer RAUC 0.3150 vs. 0.3089, and AULC 0.3829 vs. 0.3717. It also improves clean success (0.8644 vs. 0.8364), end-effector transfer RAUC (0.3143 vs. 0.3083), and RAUC AULC (0.3825 vs. 0.3709) over uniform. Clean-only has a negligible clean-success edge (0.8646 vs. 0.8644), and uniform still has a higher median r_{80} (0.5047 vs. 0.4982) in the original sweep; pooled, uniform remains higher on median r_{80} (0.5048 vs. 0.4985), so we treat this as positive manipulation-style evidence rather than final robotics evidence.

We also audited the real-simulator path as premise validation. A GPU/EGL LIBERO-Goal probe completed 75/75 valid radius rows over five tasks, three reset states, five object-perturbation radii, and four trials per radius. Fixed-policy OpenVLA audits show clean replay success of 1.0, recovery curves that drop under occlusion, blur, and shift, and BGR-boundary selection landing in the observed counterfactual-failure band $[0.25, 0.75]$ slightly more often than random-balanced selection (0.625 vs. 0.575).

The OpenVLA-OFT bridge exports 11,776 teacher-action rows and matched BGR/random TFDS datasets. The official checkpoint scores 14/15 under the same eval. In p1024, BGR and matched random both score 14/15 clean episodes; perturbations give 0.8333 vs. 0.8000 for random and 0.8167 for official. The offset-3 follow-up gives 0.8643 vs. 0.8571, but official reaches 0.8929. Pooling gives BGR 0.8550 vs. 0.8400 for random, trailing the unadapted official checkpoint at 0.8700. At p2048, BGR and random tie clean (14/15 each); original perturbations give 0.8167 vs. 0.8000 random, tying official at 0.8167. Offset-3 gives 0.8714 vs. 0.8714 random; official reaches 0.8929. Pooling p2048 gives BGR 0.8550 vs. 0.8500 random, trailing official at 0.8700; no stable official improvement.

6 Limitations

BGR requires reset or replay access, a meaningful perturbation family, and extra rollouts for boundary estimation. It targets an observed success–failure boundary, not a formal robustness certificate. The strongest evidence is controlled procedural margin expansion; the suffix simulator is positive on final RAUC, clean success, transfer RAUC, and AULC, but still trails uniform on median critical radius. OpenVLA is a learned-policy audit rather than a robotics training claim. A full robotics claim still requires action-label validation, stable optimization, clean-data scale, and multi-task LIBERO gains over both matched random selection and the official checkpoint.

7 Conclusion

BGR converts recovery-margin measurement into a replay curriculum. Synthetic and grid-margin studies show that boundary-centered replay expands directly trainable recovery curves, and the suffix simulator shows that radius coverage carries the interface to manipulation-style replay states. The OpenVLA/LIBERO audits make the learned-policy path concrete while bounding the current claim.

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Reproducibility Checklist

This checklist is filled for the current BGR submission package. The main paper is anonymous; commands, configs, per-seed CSVs, and run ledgers are in the anonymous artifact's `README.md` and `results/README.md`.

1. General Paper Structure

1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA)

yes. The method section defines replayable states and recovery curves. It also states critical radii, BGR priority, and a training-loop method box.

1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no)

yes. The paper separates controlled margin-expansion claims, robot-suffix simulator results, OpenVLA fixed-policy audits, OpenVLA-OFT adaptation diagnostics, and tabular grid-policy setup failures.

1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no)

yes. The related-work section cites curriculum learning, PLR, PAIRED/ACCEL, domain randomization, MiniGrid, LIBERO, and OpenVLA.

2. Theoretical Contributions

2.1. Does this paper make theoretical contributions? (yes/no)

yes. The paper introduces a recovery-margin formalism and states a local margin-update proposition for boundary-centered radius sampling.

If yes, please address the following points:

2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no)

yes. Proposition 1 states the local update-kernel and stochastic-dominance assumptions; the limitations section states replay-access and perturbation-family restrictions.

2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no)

yes. The local theoretical claim is stated as Proposition 1. Empirical claims are stated with measured metrics and paired-seed tests.

2.4. Proofs of all novel claims are included (yes/partial/no)

yes. Proposition 1 includes a proof sketch based on monotone coupling and monotonicity of the update kernel.

2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no)

yes. The method section gives the local margin-update intuition and the proposition proof sketch.

2.6. Appropriate citations to theoretical tools used are given (yes/partial/no)

yes. The proof cites a standard stochastic-orders reference for the quantile coupling used in Proposition 1.

2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA)

partial. The proposition is a local model; the grid ablation empirically tests its main implication that boundary-centered radius sampling matters.

2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA)

yes. Ablation scripts, configs, per-seed CSVs, and claim-checking scripts are included.

3. Dataset Usage

3.1. Does this paper rely on one or more datasets? (yes/no)

partial. Main BGR training results use procedural synthetic/grid/suffix generators. The LIBERO/OpenVLA recovery audit uses existing benchmark task definitions, init states, and fixed-policy rollout artifacts to validate learned-policy recovery-curve measurement.

If yes, please address the following points:

3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA)

yes. The paper motivates synthetic recovery curves, procedural grid-margin replay states, robot suffix diagnostics, and LIBERO/OpenVLA as a realistic manipulation-policy target.

3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA)

yes. The experiments are generated by artifact-provided scripts/configs; per-seed result CSVs are included under `results/`.

3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA)

yes. The generated procedural data and code are included in the anonymous artifact under an MIT license.

3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA)

yes. LIBERO and OpenVLA are cited; MiniGrid is cited for grid-style context.

3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA)

yes for LIBERO task definitions, benchmark assets, and OpenVLA model code/weights. The anonymous artifact contains compact derived recovery summaries; raw fixed-policy rollout logs are retained for audit but are not required for the claimed quantitative results.

3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing (yes/partial/no/NA)

NA. No non-public dataset is used for claimed quantitative results.

4. Computational Experiments

4.1. Does this paper include computational experiments? (yes/no)

yes.

If yes, please address the following points:

4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA)

partial. Final configs and ablation variants are included in the anonymous artifact; the full search history is not exhaustively enumerated in the main paper.

4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no)

yes. Procedural generation, LIBERO probing, OpenVLA recovery summarization, aggregation, and figure scripts are included.

4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no)

yes. Experiment scripts, configs, tests, aggregation, and result CSVs are included in the anonymous artifact.

4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no)

yes. The anonymous artifact includes an MIT license, and the code plus generated procedural data will be made publicly available upon publication.

4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no)

partial. Code is modular and tested; comments are intentionally sparse and do not yet cross-reference paper sections.

4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA)

yes. Every config lists seeds; scripts pass seeds to NumPy generators and result CSVs include per-seed rows.

4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no)

yes. The run ledger records Slurm partitions, CPU/memory/time requests, hostnames for logged runs, environment constraints, and JSON environment snapshots for compute and GPU jobs.

4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no)

yes. Recovery AUC, critical radius, clean success, transfer RAUC, and AULC are defined or described.

4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no)

yes. Runs are reported in the paper and run ledger: synthetic training, estimator validation, grid-margin full-baseline, grid ablation, grid sensitivity confirmations, and robot-suffix coverage comparisons use 30 paired seeds; grid learning-curve diagnostics use 15 seed pairs. Smaller three- or five-seed exploratory diagnostics are labeled as diagnostics in the run ledger rather than used as main claims.

4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no)

yes. Aggregated tables include standard errors; experiments report median critical radius and AULC in addition to final means.

4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no)

yes. The anonymous artifact includes paired exact sign tests for the reported paired comparisons, including the 30-seed suffix coverage confirmation, grid-margin target-sensitivity, grid-margin obstacle-regime sensitivity, grid-margin geometry-stress sensitivity, and grid-margin learning-curve sweeps. The paper separates these main comparisons from smaller exploratory evidence.

4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA)

yes. Final hyperparameters are in the artifact YAML configs under `configs/`.